Youssef Naciri Zerta

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Fès (Morocco) until February 2027 | Cambridge (UK) February–August 2027

Research Interests

I want to understand how to make cellular behavior reliably controllable, particularly how to move cells between states and eventually design new cellular functions. This means understanding how regulatory mechanisms give rise to cell states, what our measurements and models can actually tell us about those mechanisms, and how to use that knowledge to design interventions that achieve desired outcomes, including in conditions we haven't tested before. My work spans experimental systems biology, causal representation learning, mechanistic modeling, perturbation genomics, and scientific software engineering.

Research Experience

Research Thesis: Toward Predictive Control of Cell State 2026–2027

EMBL-EBI, Saez Lab, with the Wellcome Sanger Institute — with Rory Maizels – *research project 09/2026–02/2027 (remote, part time); thesis 02–08/2027 (full time, on site)*

- **Intervention-relevant identifiability (ongoing):** Simulate published gene-regulatory networks with ordinary differential equations and realistic count noise. Compare knockdowns, knockouts, and calibrated state-dependent transcription-factor inputs to determine which perturbations distinguish mechanisms *and* whether remaining ambiguity changes intervention choice.
- **Perturbation generalization (planned thesis component):** Use differentiating stem-cell perturbation data to learn causal representations of regulatory state and evaluate predictions on held-out interventions, with the longer-term aim of selecting interventions toward a target state.

AnnNet and SysBioVerse — Lead Developer 2025–Present

EMBL-EBI, Saez Lab, with Open Targets – *co-first-author manuscript in preparation*

- Designed AnnNet's incidence representation, annotation model, storage format, and application interfaces; supports directed, signed, hyperedge, multilayer, and condition-specific networks with lossless storage and interoperability with NetworkX, igraph, graph-tool, and PyTorch Geometric.
- AnnNet is used by groups at EMBL-EBI and Imperial College London and developed with scverse core developers for the single-cell Python ecosystem.
- Extending the infrastructure into SysBioVerse (2026–), a consortium/interface for regulatory, signalling, metabolic, cell–cell, and multi-omic causal-inference tools.

Sparsity, Not Alignment: How a Standard Interpretability Test Manufactures Discovery 2026

Sole author – *accepted, NeurIPS 2026 workshop; paper*

- Tested whether sparse-autoencoder features of a genomic foundation model correspond to regulatory mechanism, using saturation-mutagenesis MPRA (massively parallel reporter assay) data as per-base ground truth. Run the standard way, the test reports a discovery that passes every control the literature uses: label shuffling, repeat masking, a likelihood baseline, three seeds, two autoencoder families and two architectures.
- The discovery is an artifact. The test thresholds one tail of a null distribution, and the same FDR (false discovery rate) procedure applied to the other tail finds more "discoveries". The two distributions differ in spread, not location, and the effect grows with dictionary size, because sparser units have noisier scores.
- Proposed four diagnostics that cost nothing extra, a studentised repair, and a noise ceiling showing that the accompanying reproducibility failure comes from the size of the ground-truth dataset, not from the decomposition.

Multiplex Bacterial Interaction Networks 2024–Present

Université Euromed de Fès – *experimental systems biology; first author; preprint, under review at ISME Communications*

- Designed and ran a trait-resolved interaction study across 60 soil bacterial strains: pairwise assays producing 12 directed interaction layers, analysed as one multilayer network rather than a single graph.
- Showed that the sign and strength of an interaction depend on context rather than being fixed for a pair of strains, and placed the strains in a low-dimensional space of antagonistic, metabolic and morphogenetic strategies. This representational problem is what led to AnnNet.

SampleCLR — Patient-Level Representations from Single-Cell Transcriptomics 2025

Helmholtz Munich, Lücken Lab – *contrastive and attention-based sample representation; preprint*

- Contributed the interpretability and transfer-learning analyses and parts of the architecture design for a model that learns patient-level representations from single-cell data; the paper reports state-of-the-art disease-stratification performance.

- Tested whether the representations transfer across biological systems rather than only fitting the training benchmark.

Also: analysis pipeline for detecting biomarker combinations from epigenomic data, MD Anderson Cancer Center / KAUST, 2024–2025.

Publications, Preprints & Manuscripts

- **Y. Zerta**, et al. *Multilayer interaction networks reveal multidimensional ecological strategies and context-dependent competition in soil actinobacteria* First author. bioRxiv (2026), doi:10.64898/2026.09.16.752070. Under review at *ISME Communications*.
- **Y. Naciri Zerta.** *Sparsity, not alignment: how a standard interpretability test manufactures discovery in a genomic foundation model.* Sole author. Accepted, NeurIPS 2026 workshop. paper.
- *Great patients embed alike: contrastive learning for sample representation from single-cell data.* Middle author. preprint.
- *AnnNet: a unified annotated-graph structure for biological networks.* Co-first author; **manuscript in preparation** (not yet a public preprint).

Education

Ingénieur d'État en Biotechnologies (State Engineer, Biotechnology) Expected 2027

Université Euromed de Fès (UEMF), Morocco – *five-year engineering degree; Master's-equivalent, 300 ECTS (European Credit Transfer and Accumulation System)*

Relevant coursework: Systems Biology, Synthetic Microbiology, Recombinant DNA Technology, Statistical Genetics, Analytic Chemistry, Organic Chemistry, Probability & Statistics.

Methods

Causal inference & representation learning: intervention-relevant identifiability, perturbation generalization, causal representations, sparse autoencoders, contrastive learning, mechanistic interpretability.

Modelling & statistics: ordinary differential equation models of gene regulation, Bayesian model fitting, high-dimensional and sparse inference.

Evaluation design: interventional ground truth, null calibration, permutation inference, multiple testing, group-structured cross-validation, leakage diagnosis, noise ceilings, planted nulls.

Genomics & regulation: gene regulatory networks, perturbational genomics, single-cell transcriptomics, MPRA data, statistical genetics.

Systems biology: mechanistic and dynamical modelling, multilayer and hypergraph formalisms, prior-knowledge integration.

Engineering: Python (PyTorch, NumPy, scikit-learn), scientific library design, open-source maintenance, GPU training, cloud computing, Docker.

Wet lab: molecular cloning, recombinant DNA, genome engineering, high-throughput microbial culture and interaction assays, analytical chemistry.

Languages: English & French (fluent), Arabic & Tamazight (native), Spanish (basic).